

The Language of Medicine

13th edition

Davi-Allen Chabner

Chapter 18

Endocrine System

Chapter Goals (Slide 1 of 2)

- Identify the endocrine glands and their hormones.
- Gain an understanding of the functions of these hormones in the body.
- Analyze medical terms related to the endocrine glands and their hormones.

Chapter Goals (Slide 2 of 2)

- Identify abnormal conditions resulting from excessive and deficient secretions of the endocrine glands.
- Describe laboratory tests and clinical procedures related to endocrinology, and recognize relevant abbreviations.
- Apply your new knowledge to understanding medical terms in their proper contexts, such as medical reports and records.

Chapter 18

Lesson 18.1

Introduction

Endocrine System (Slide 1 of 2)

- Glands release hormones.
- Hormones regulate the many and varied functions of an organism.
- Hormones bind to receptors.
- Receptors are recognition sites in the various target tissues on which hormones act.

Introduction

Endocrine System (Slide 2 of 2)

- **ENDOCRINE** glands—
Secrete their hormones directly into the bloodstream
- **EXOCRINE** glands—
Send chemical substances (tears, sweat, milk, saliva) via ducts to the outside of the body

Endocrine Glands

- Thyroid gland
- Parathyroid glands (four glands)
- Adrenal glands (one pair)
- Pancreas (islets of Langerhans)
- Pituitary gland

- Ovaries (one pair)
- Testes (one pair)
- Pineal gland

The Endocrine System

- Exocrine glands include the following:
- Thyroid gland
- Parathyroid glands (four glands)
- Adrenal glands (one pair)
- Pancreas (islet of Langerhans)
- Pituitary gland
- Ovaries in female (one pair)
- Testes in male (one pair)
- Pineal gland

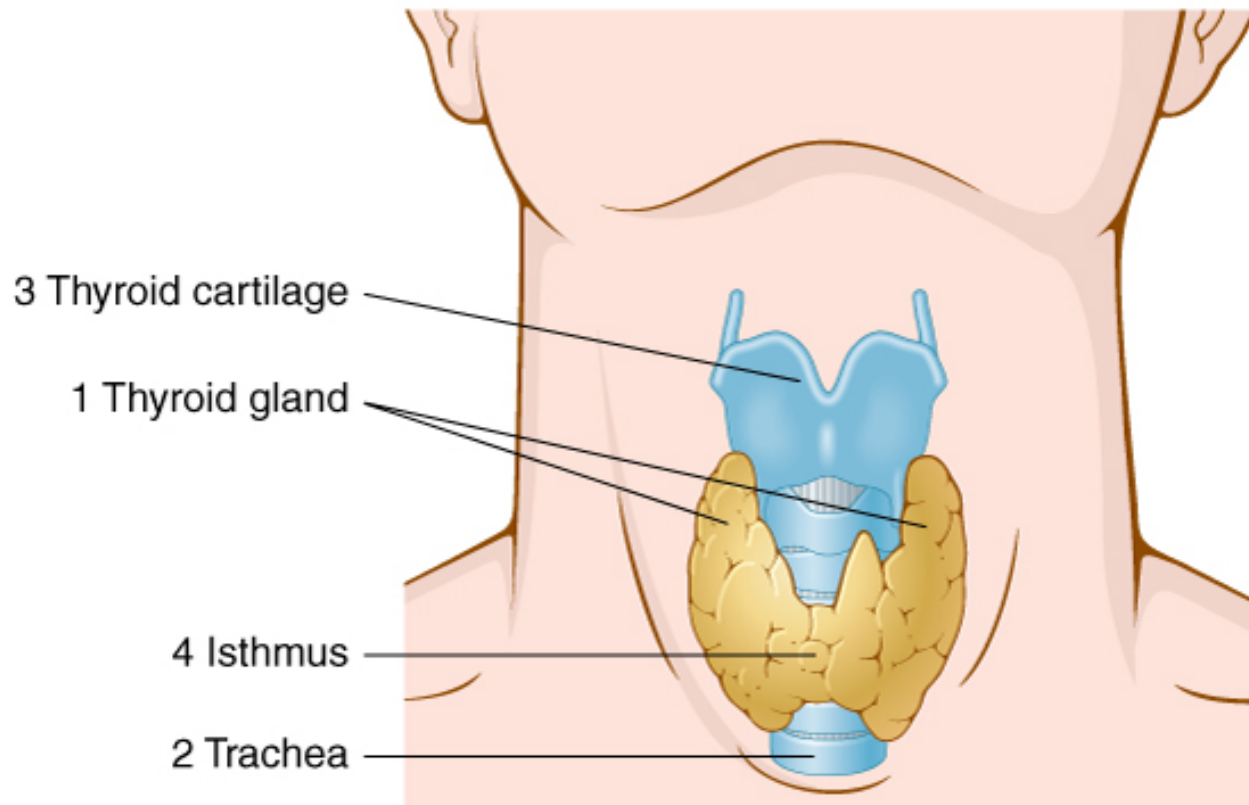
Endocrine Tissue (Not including major glands)

TABLE 18-1 ENDOCRINE TISSUE (APART FROM MAJOR GLANDS): LOCATION, SECRETION, AND ACTION		
LOCATION	SECRETION	ACTION(S)
Body cells	Prostaglandins	Aggregation of platelets Contract uterus Lower acid secretion in stomach Lower blood pressure
Gastrointestinal tract	Cholecystokinin Gastrin Secretin	Contracts gallbladder Stimulates gastric secretion Stimulates pancreatic enzymes
Kidney	Erythropoietin	Stimulates erythrocyte production
Placenta	Human chorionic gonadotropin	Sustains pregnancy
Skin	Vitamin D	Affects absorption of calcium

Thyroid Gland (Slide 1 of 2)

- Anatomical structures of the thyroid gland:
- Thyroid
- Trachea
- Thyroid cartilage
- Isthmus

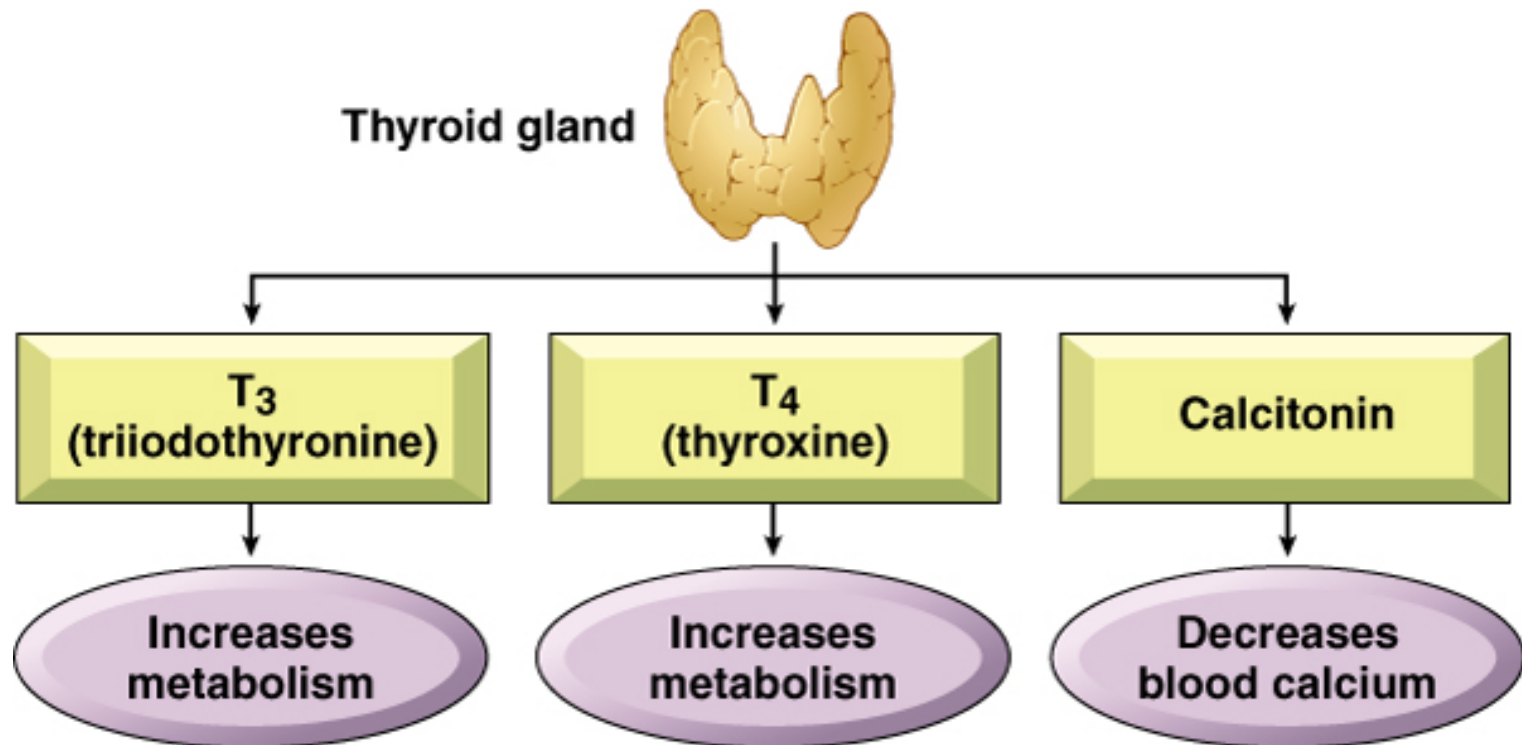
Thyroid Gland (Slide 2 of 2)



Thyroid Function

- There are two hormones:
 - Thyroxine or tetraiodothyronine (T_4)
 - Triiodothyronine (T_3).
- Thyroid hormones aid cells in their uptake of oxygen and regulate metabolic rate.
- **Calcitonin:** stimulates calcium to leave the blood and enter the bone.

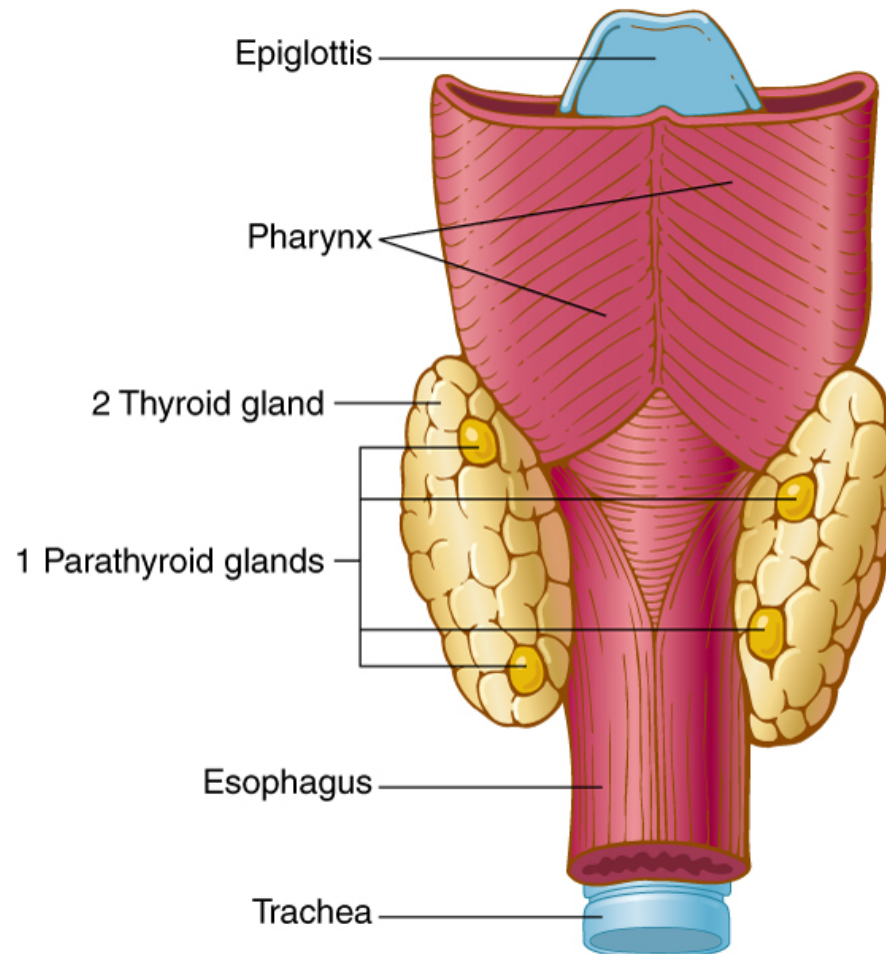
Thyroid Gland: Hormones and Actions



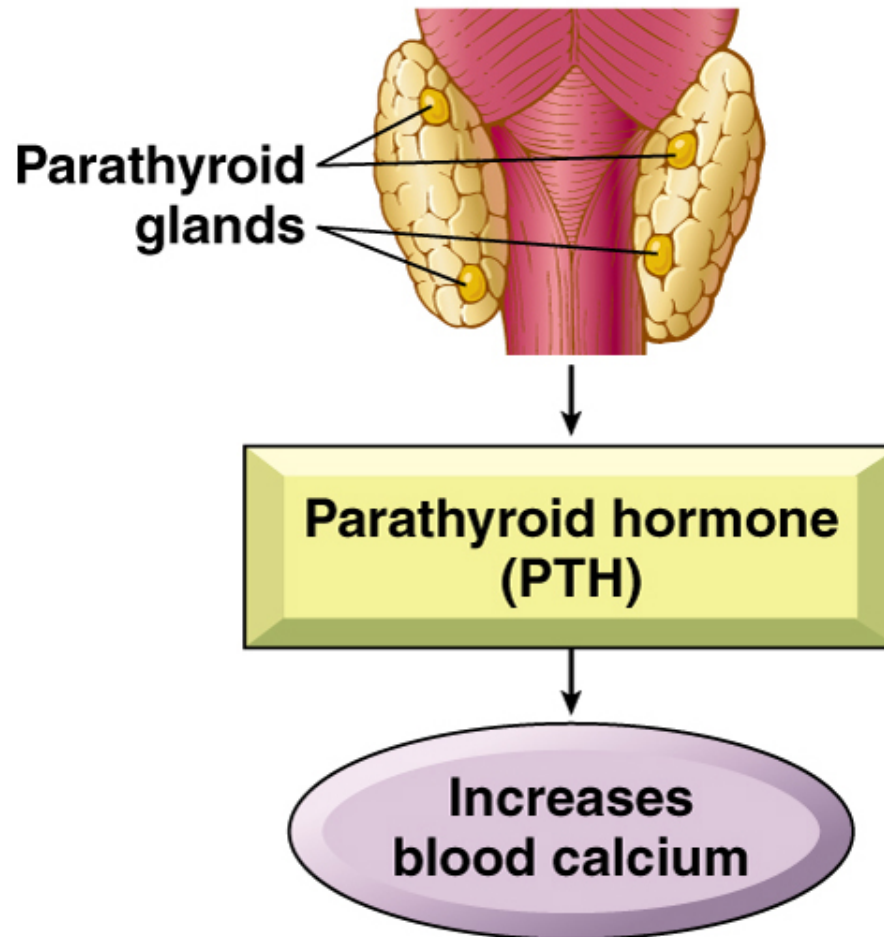
Parathyroid Glands (Slide 1 of 3)

- Parathyroid glands are four small oval bodies located on the dorsal aspect of the thyroid gland.

Parathyroid Glands (Slide 2 of 3)



Parathyroid Glands (slide 3 of 3)



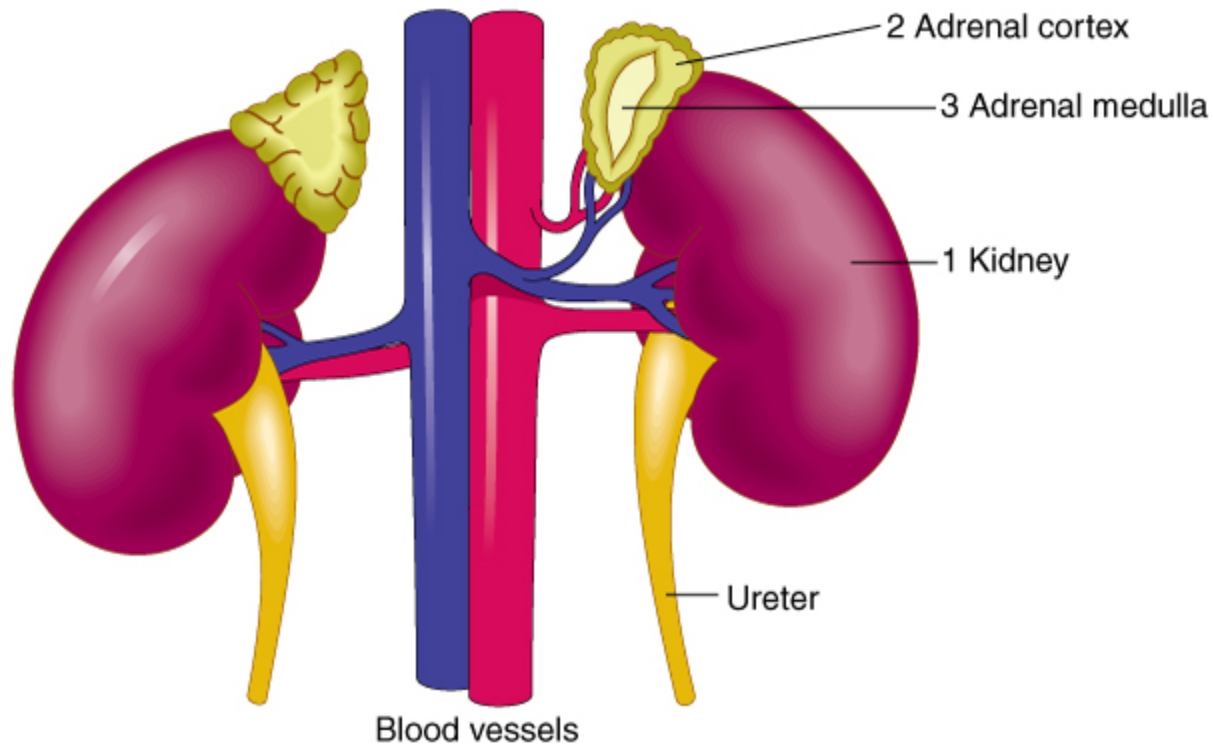
Parathyroid Function

- **Parathyroid hormone (PTH):** causes calcium to mobilize from bones into the bloodstream

Adrenal Glands (Slide 1 of 2)

- Adrenal glands are located on top of each kidney. Each gland consists of two parts:
- Adrenal cortex (outer portion)
- Adrenal medulla (inner portion)

Adrenal Glands (Slide 2 of 2)



Adrenal Glands:

Location and Structure

Each gland has two parts

- An **outer** portion, the **adrenal cortex**
 - Secretes corticosteroids or steroids, chemicals derived from cholesterol

- An **inner** portion, **adrenal medulla**
 - Secretes catecholamines
chemicals derived from amino acids

Function:

Adrenal Cortex Secretes

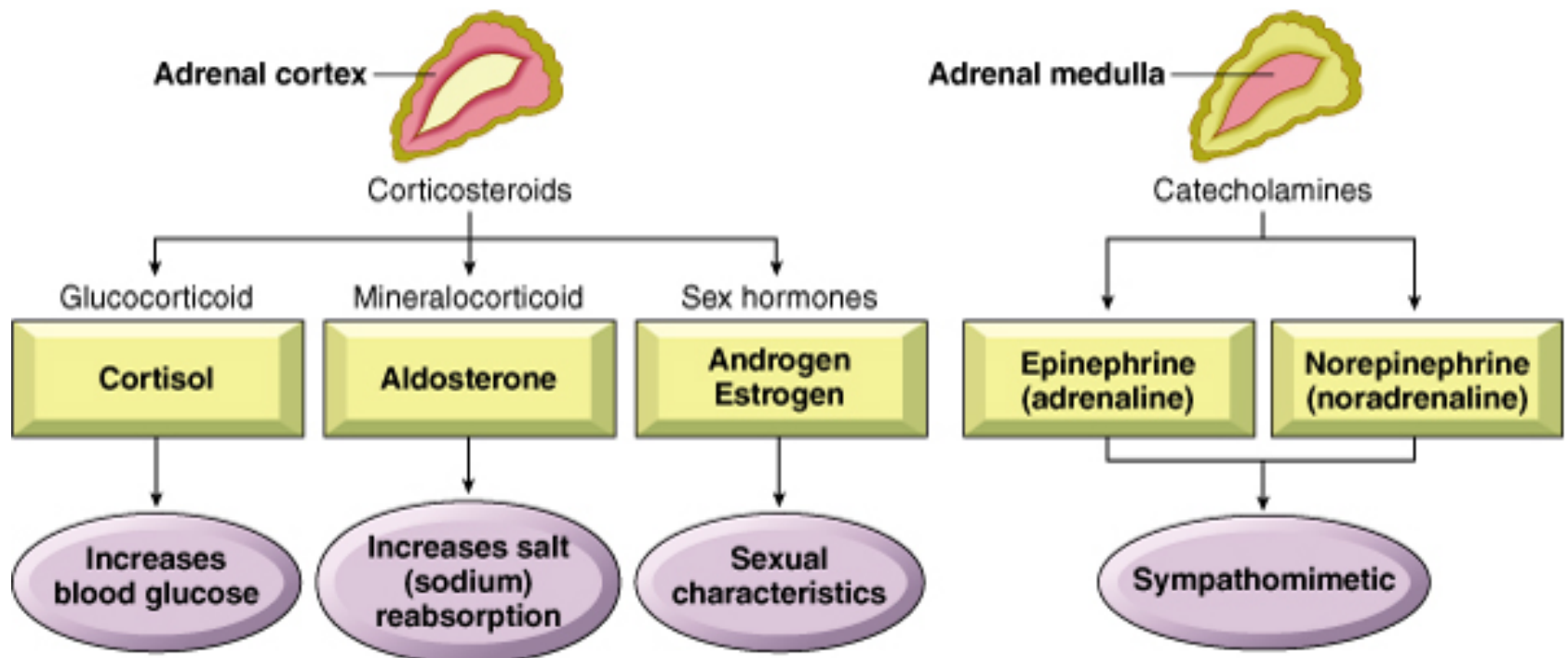
- **Glucocorticoids:** influence metabolism of sugars, fats, and proteins (**cortisol**) and are anti-inflammatory (**cortisone**). Influences—**SUGAR**
- **Mineralocorticoids:** regulate electrolytes
 - **Aldosterone:** reabsorption of sodium/excretion of potassium. Influences—**SALT**
- **Sex Hormones:** androgens (testosterone) and estrogens influence secondary sex characteristics

Function:

Adrenal Medulla Secretes

- Two types of **catecholamine** hormones
 - **Epinephrine (adrenaline)**: increases heart rate and blood pressure, dilates bronchial tubes, releases glucose from storage
 - **Norepinephrine (noradrenaline)**: constricts vessels to raise blood pressure
 - **Both** are sympathomimetic agents.

The Adrenal Cortex and Adrenal Medulla



Pancreas

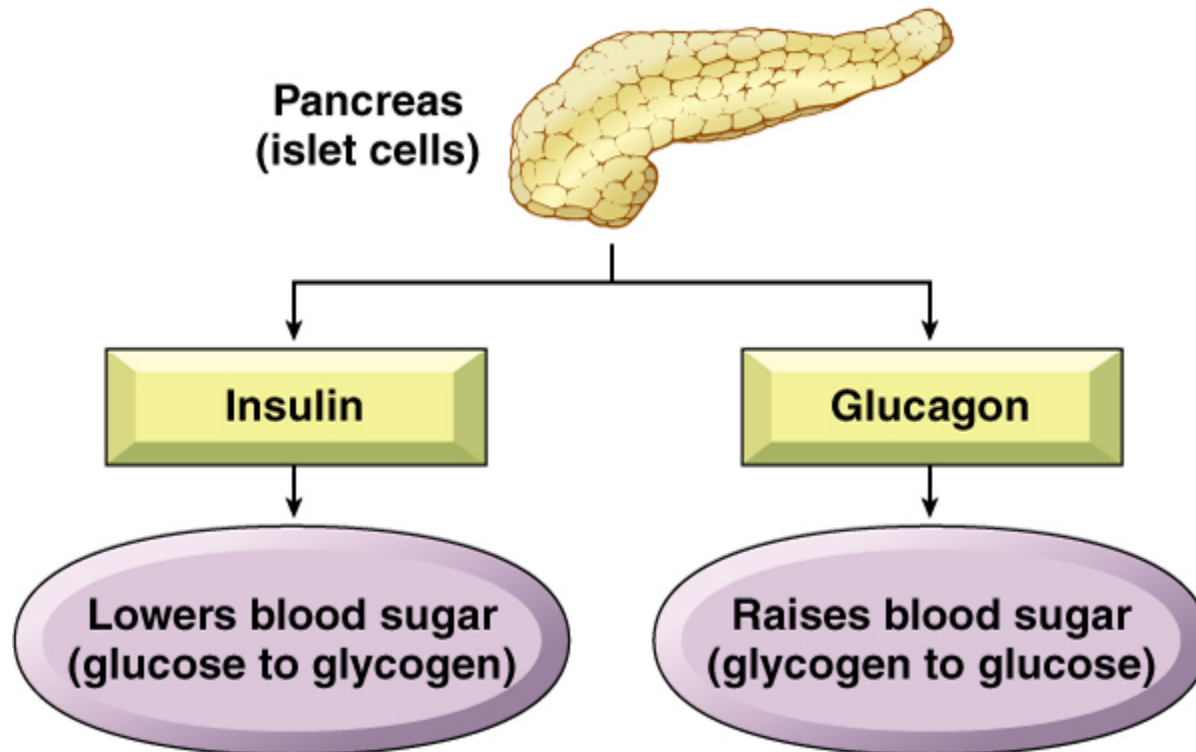
- Located near and partially behind the stomach
- Exocrine and endocrine organ

Pancreas Function (Slide 1 of 2)

Endocrine function

- Islets of Langerhans produce:
 - **Insulin:** promotes movement of glucose into cells and promotes storage as glycogen
 - **Glucagon:** promotes movement of glucose into the blood by breaking down glycogen stored in liver cells

Pancreas Function (Slide 2 of 2)



Chapter 18

Lesson 18.2

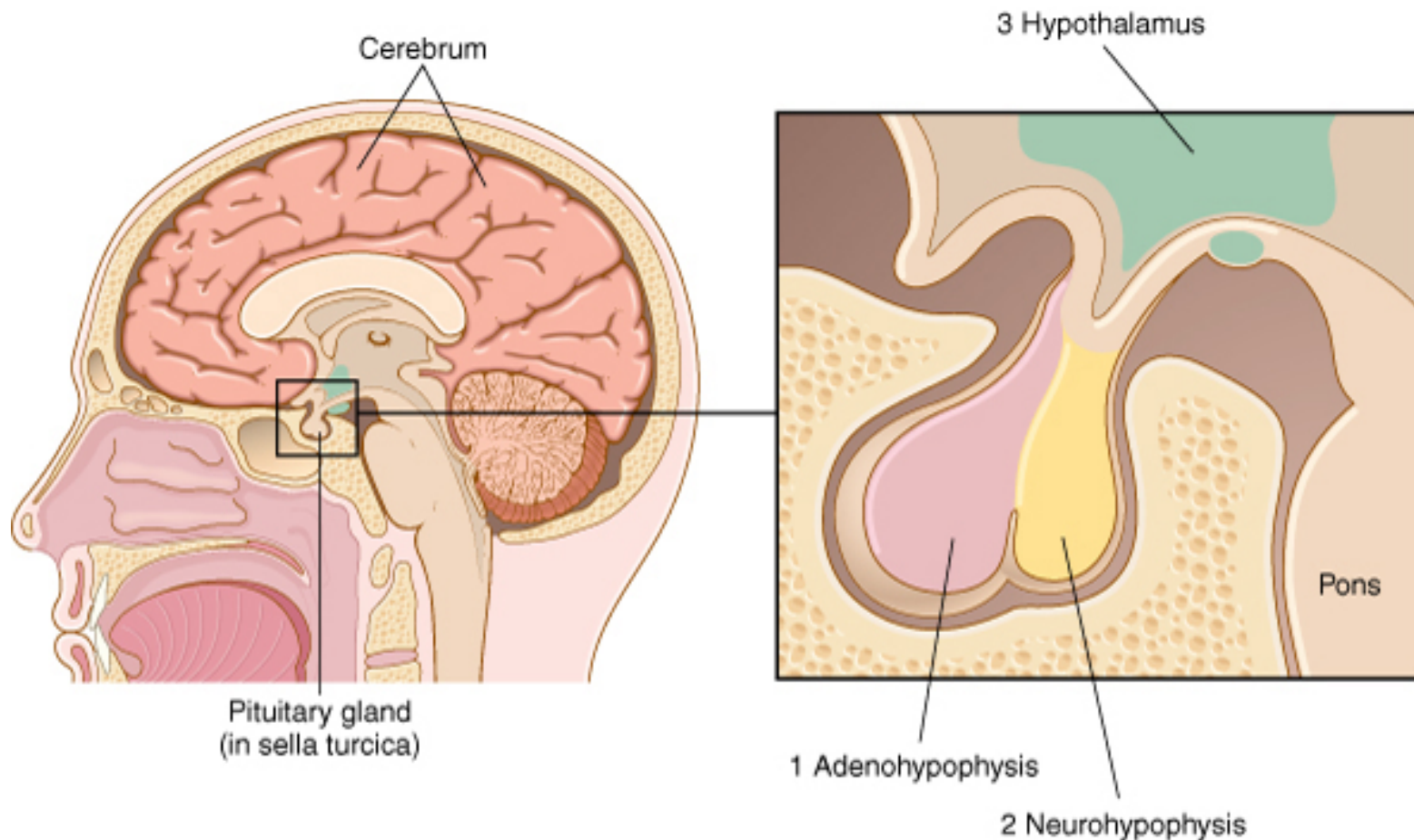
Pituitary Gland (Slide 1 of 3)

- Pea-sized gland in depression of skull (sella turcica) also called the **hypophysis**
 - Anterior lobe (adenohypophysis)
 - Posterior lobe (neurohypophysis)
- **Hypothalamus** controls secretions of the pituitary via releasing factors (hormones)

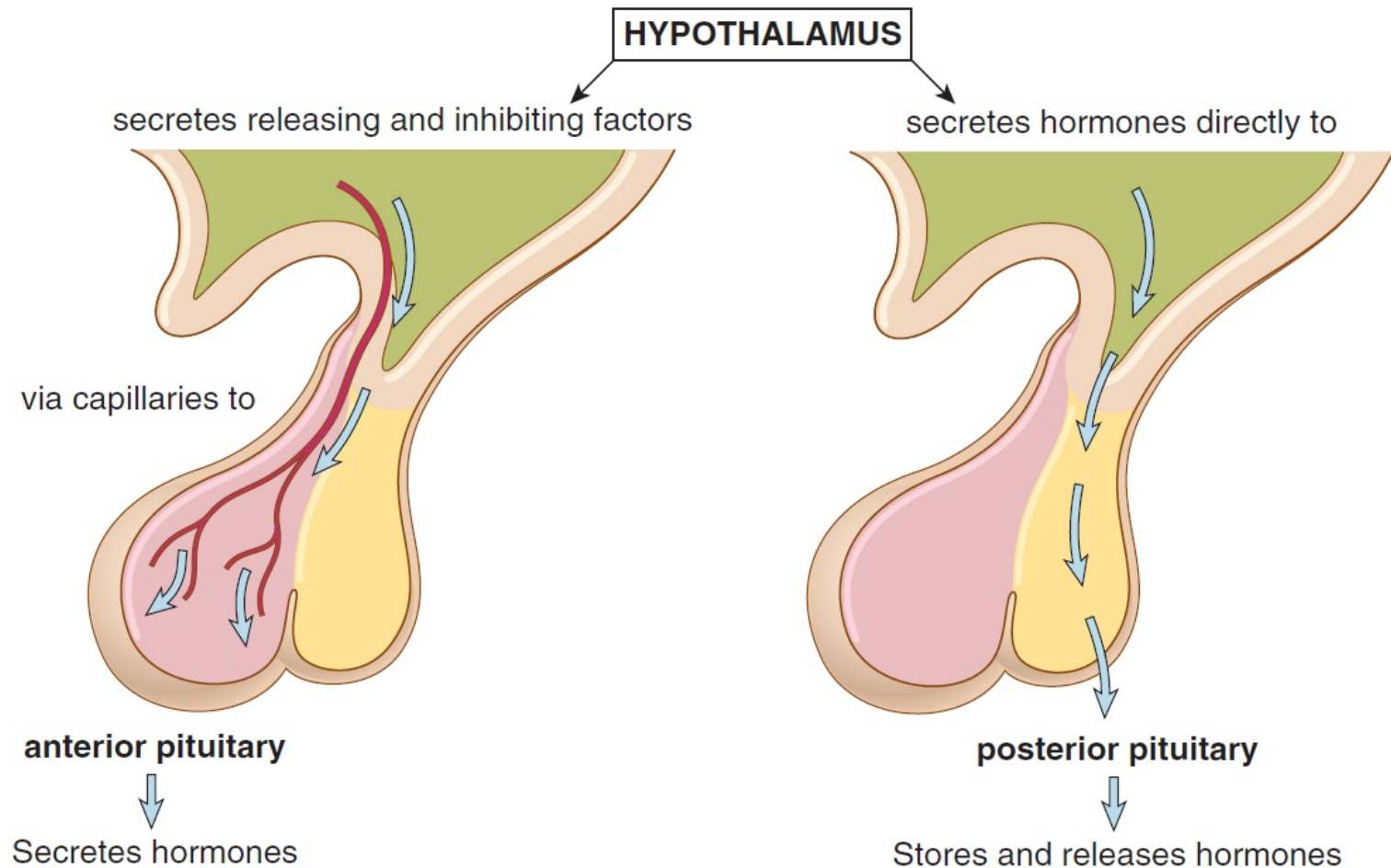
Pituitary Gland (Slide 2 of 3)

- The pituitary gland consists of two distinct parts:
- Anterior lobe or adenohypophysis
- Posterior lobe or neurohypophysis

Pituitary Gland (Slide 3 of 3)



Relationship of the Hypothalamus to the Pituitary Gland



Pituitary Function (Slide 1 of 2)

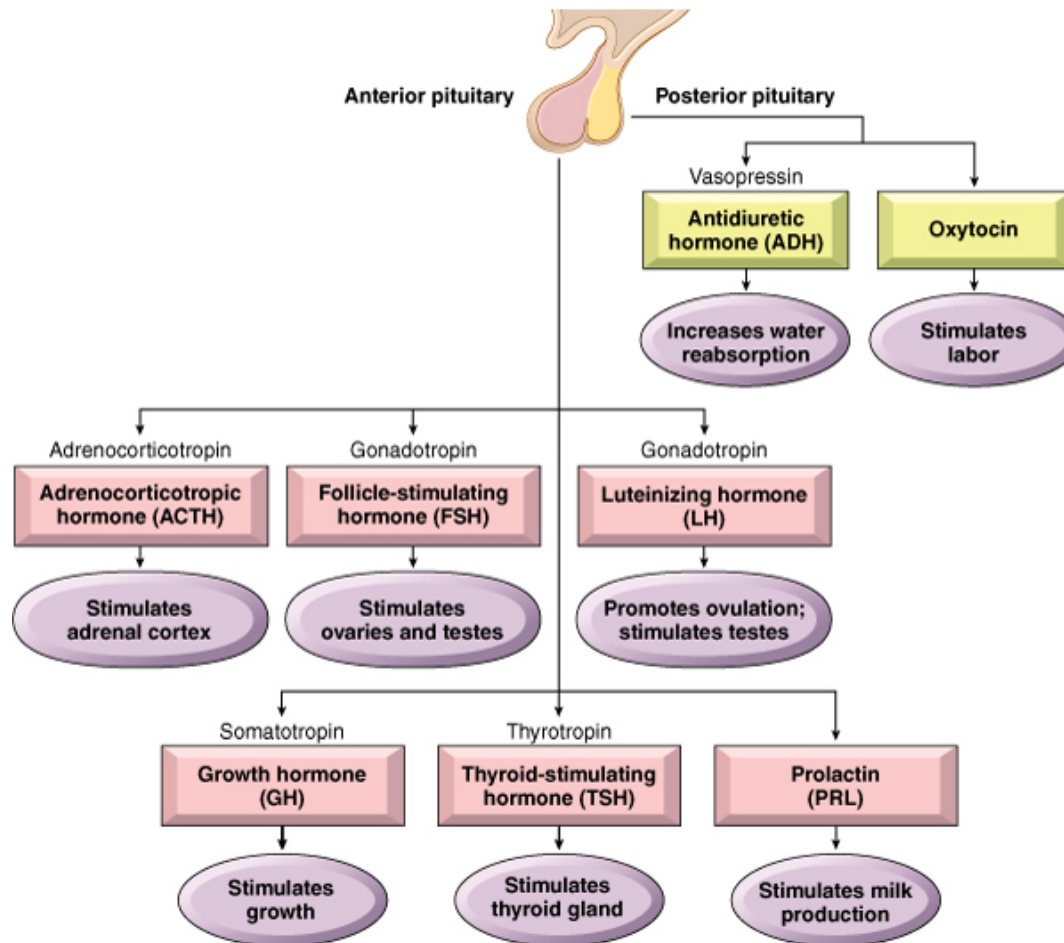
- **Anterior pituitary**

- Growth hormone (GH) or somatotropin (STH)
- Thyroid-stimulating hormone (TSH; thyrotropin)
- Adrenocorticotrophic hormone (ACTH)
- Gonadotropic hormones (FSH, LH)
- Prolactin (PRL)

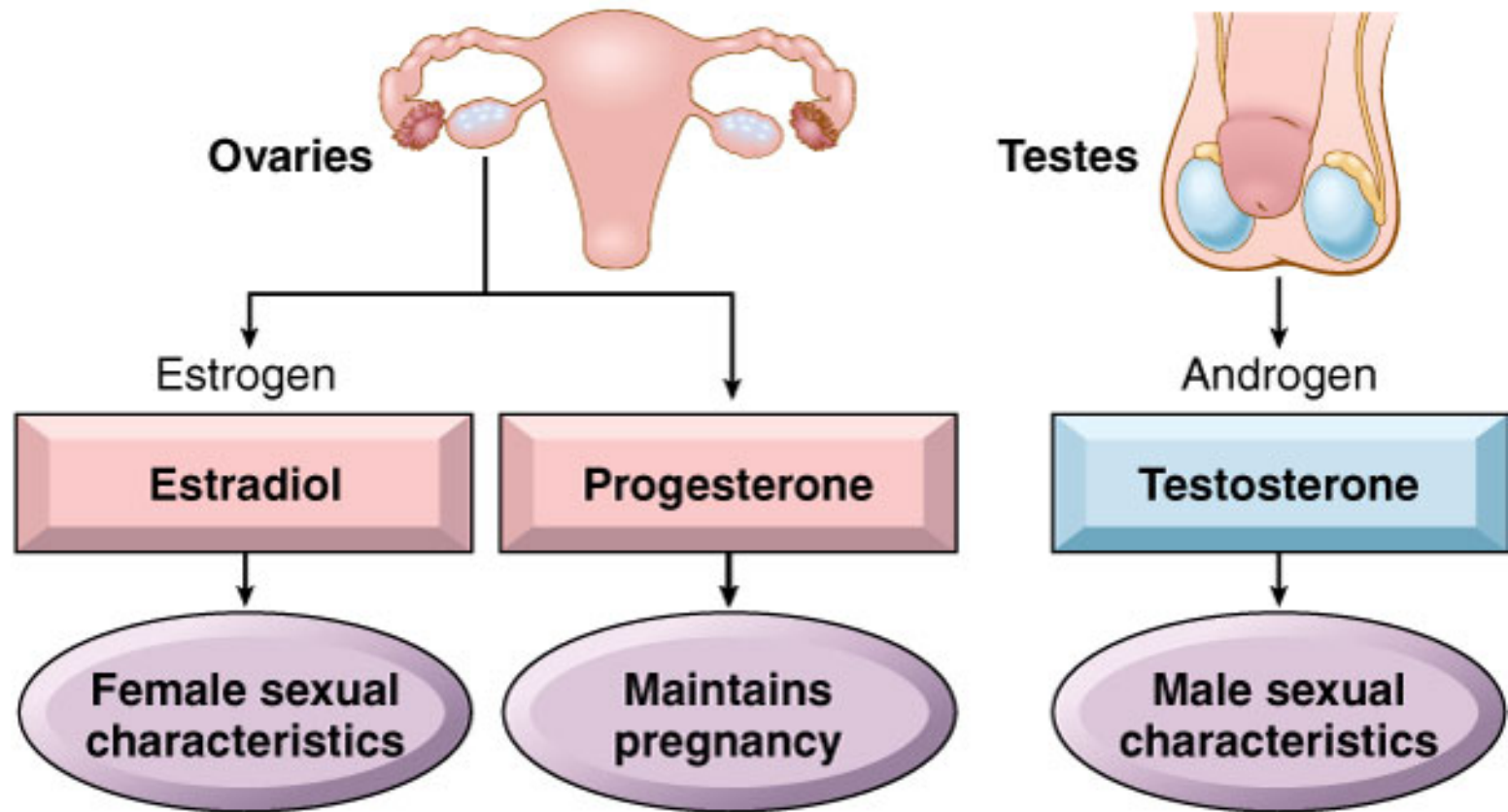
Pituitary Function (Slide 2 of 2)

- **Posterior pituitary:** stores and releases hormones synthesized in the hypothalamus
 - **Antidiuretic hormone (ADH; vasopressin)**
 - **Oxytocin (OT)**

Pituitary Gland: Hormones and Actions



Ovaries and Testes: Hormones and Actions



Male

Testes

- Hormone:
testosterone
Maintains germ cell formation and secondary sexual characteristics

Female

Ovaries

- Hormones: **estrogen** and **progesterone**
Maintains menstrual cycle, release of ovum, secondary sexual characteristics, preparation of uterus for pregnancy

Vocabulary – Major Endocrine Glands

(Slide 1 of 4)

Term
adrenal cortex
adrenal medulla
ovaries
pancreas

Vocabulary – Major Endocrine Glands

(Slide 2 of 4)

Term	Meaning/Definition
adrenal cortex	Outer section of each adrenal gland; secretes cortisol, aldosterone, and sex hormones
adrenal medulla	Inner section of each adrenal gland; secretes epinephrine and norepinephrine
ovaries	Located in the lower abdomen of female; responsible for egg production and estrogen and progesterone secretion
pancreas	Located behind the stomach

Vocabulary – Major Endocrine Glands

(Slide 3 of 4)

Term
parathyroid glands
pituitary gland (hypophysis)
testes
thyroid gland

Vocabulary – Major Endocrine Glands

(Slide 4 of 4)

Term	Meaning/Definition
parathyroid glands	Four small glands on the posterior of the thyroid gland
pituitary gland (hypophysis)	Located at the base of brain in the sella turcica
testes	Two glands enclosed in the scrotal sac of a male; responsible for sperm production and testosterone secretion
thyroid gland	Located in the neck on either side of the trachea; secretes thyroxine, triiodothyronine, and calcitonin

Vocabulary – Hormones (Slide 1 of 14)

Term
adrenaline (epinephrine)
adrenocortico- tropic hormone (ACTH)
aldosterone
androgen

Vocabulary – Hormones (Slide 2 of 14)

Term	Meaning/Definition
adrenaline (epinephrine)	Secreted by the adrenal medulla; increases heart rate and blood pressure
adrenocortico- tropic hormone (ACTH)	Secreted by the anterior lobe of the pituitary gland; stimulates adrenal cortex
aldosterone	Secreted by the adrenal cortex; increases salt reabsorption
androgen	Male hormone secreted by the testes and adrenal cortex (to a lesser extent)

Vocabulary – Hormones (Slide 3 of 14)

Term
antidiuretic hormone (ADH)
calcitonin
cortisol
epinephrine (adrenaline)

Vocabulary – Hormones (Slide 4 of 14)

Term	Meaning/Definition
antidiuretic hormone (ADH)	Secreted by the posterior lobe of the pituitary gland; increases reabsorption of water by kidney
calcitonin	Secreted by the thyroid gland; decreases blood calcium levels
cortisol	Secreted by the adrenal cortex; increases blood sugar
epinephrine (adrenaline)	Secreted by the adrenal medulla; increases heart rate and blood pressure

Vocabulary – Hormones (Slide 5 of 14)

Term
estradiol
estrogen
follicle- stimulating hormone (FSH)
glucagons

Vocabulary – Hormones (Slide 6 of 14)

Term	Meaning/Definition
estradiol	Estrogen secreted by the ovaries
estrogen	Female hormone secreted by the ovaries and adrenal cortex (to lesser extent)
follicle-stimulating hormone (FSH)	Secreted by the anterior lobe of the pituitary gland; stimulates hormone secretion and egg production by ovaries and sperm production by testes
glucagons	Secreted by alpha islet cells of pancreas; increases blood sugar

Vocabulary – Hormones (Slide 7 of 14)

Term
growth hormone (GH); somatotropin
insulin
luteinizing hormone (LH)

Vocabulary – Hormones (Slide 8 of 14)

Term	Meaning/Definition
growth hormone (GH); somatotropin	Secreted by the anterior lobe of the pituitary gland; stimulates growth of bones and soft tissues
insulin	Secreted by beta islet cells of pancreas; helps glucose pass into cells and promotes conversion of glucose to glycogen
luteinizing hormone (LH)	Secreted by the anterior lobe of pituitary gland; stimulates ovulation in females and testosterone secretion in males

Vocabulary – Hormones (Slide 9 of 14)

Term
norepinephrine
oxytocin (OT)
parathormone (PTH)
progesterone

Vocabulary – Hormones (Slide 10 of 14)

Term	Meaning/Definition
norepinephrine	Secreted by the adrenal medulla; increases heart rate and blood pressure
oxytocin (OT)	Secreted by the posterior lobe of the pituitary gland; stimulates contraction of uterus during labor and childbirth
parathormone (PTH)	Secreted by parathyroid glands; increases blood calcium
progesterone	Secreted by the ovaries; prepares the uterus for pregnancy

Vocabulary – Hormones (Slide 11 of 14)

Term
prolactin (PRL)
somatotropin (STH)
testosterone
thyroid-stimulating hormone (TSH); thyrotropin

Vocabulary – Hormones (Slide 12 of 14)

Term	Meaning/Definition
prolactin (PRL)	Secreted by the anterior lobe of the pituitary gland; promotes milk secretion
somatotropin (STH)	Secreted by the anterior lobe of the pituitary gland; growth hormone
testosterone	Male hormone secreted by the testes
thyroid-stimulating hormone (TSH); thyrotropin	Secreted by the anterior lobe of the pituitary gland; promotes thyroid function

Vocabulary – Hormones (Slide 13 of 14)

Term
thyroxine (T_4)
triiodothyronine (T_3)
vasopressin

Vocabulary – Hormones (Slide 14 of 14)

Term	Meaning/Definition
thyroxine (T_4)	Secreted by the thyroid gland; increases cell metabolism
triiodothyronine (T_3)	Secreted by the thyroid gland; increases cell metabolism
vasopressin	Secreted by the posterior lobe of the pituitary gland; antidiuretic hormone (ADH) ; raises water reabsorption and raises blood pressure

Vocabulary – Related Terms (Slide 1 of 8)

Term
catecholamines
corticosteroids
electrolyte
glucocorticoid

Vocabulary – Related Terms (Slide 2 of 8)

Term	Meaning/Definition
catecholamines	Hormones derived from an amino acid and secreted by the adrenal medulla
corticosteroids	Hormones produced by the adrenal cortex
electrolyte	Mineral salt found in blood and tissues
glucocorticoid	Steroid hormone secreted by the adrenal cortex; regulates glucose, fat, and protein metabolism

Vocabulary – Related Terms (Slide 3 of 8)

Term
homeostasis
hormone
hypothalamus
mineralocorticoid

Vocabulary – Related Terms (Slide 4 of 8)

Term	Meaning/Definition
homeostasis	Tendency of an organism to maintain a constant internal environment
hormone	Substance secreted by an endocrine gland to influence structure or function of an organ or gland
hypothalamus	Region of the brain lying below the thalamus and above the pituitary gland
mineralocorticoid	Steroid hormone secreted by the adrenal cortex to regulate mineral salts and water balance in the body

Vocabulary – Related Terms (Slide 5 of 8)

Term
receptor
sella turcica
sex hormones

Vocabulary – Related Terms (Slide 6 of 8)

Term	Meaning/Definition
receptor	Cellular or nuclear protein that binds to a hormone so that a response can be elicited
sella turcica	Cavity in the skull that contains the pituitary gland
sex hormones	Steroids produced by the adrenal cortex to influence male and female sexual characteristics

Vocabulary – Related Terms (Slide 7 of 8)

Term
steroid
sympatho- mimetic
target tissue

Vocabulary – Related Terms (Slide 8 of 8)

Term	Meaning/Definition
steroid	Complex substance related to fats and of which many hormones are made
sympatho-mimetic	Pertaining to mimicking or copying the effect of the sympathetic nervous system
target tissue	Cells of an organ that are affected or stimulated by specific hormones

QUICK QUIZ (Slide 1 of 5)

1. The outer section of each adrenal gland is the:

- A. adrenal medulla
- B. adrenal cortex
- C. adrenaline
- D. adrenocorticotrophic hormone

QUICK QUIZ (Slide 2 of 5)

2. Secreted by the anterior lobe of the pituitary gland; stimulates ovulation in females and testosterone secretion in males:

- A. luteinizing hormone (LH)
- B. estrodoil
- C. oxytocin
- D. prolactin

Combining Forms, Suffixes, Prefixes, and Terminology – Glands (Slide 1 of 2)

Combining Form	Meaning
aden/o	gland
adrenal/o	adrenal glands
gonad/o	sex glands (ovaries, testes)
pancreat/o	pancreas
parathyroid/o	parathyroid gland

Combining Forms, Suffixes, Prefixes, and Terminology – Glands (Slide 2 of 2)

Combining Form	Meaning
pituitar/o	pituitary gland; hypophysis
thyr/o	thyroid gland
thyroid/o	thyroid gland

Combining Forms, Suffixes, Prefixes, and Terminology – Related Forms (Slide 1 of 3)

Combining Form	Meaning
andr/o	male
calc/o, calici/o	calcium
cortic/o	cortex, outer region
crin/o	secrete
dips/o	thirst
estr/o	female
gluc/o, glyc/o	sugar

Combining Forms, Suffixes, Prefixes, and Terminology – Glands (Slide 2 of 3)

Combining Form	Meaning
home/o	sameness
hormon/o	hormone
kal/I	potassium (an electrolyte)
lact/o	milk
myx/o	mucus
natr/o	sodium (an electrolyte)

Combining Forms, Suffixes, Prefixes, and Terminology – Related Forms

(Slide 3 of 3)

Combining Form	Meaning
phys/o	growing
somat/o	body
ster/o	solid structure
toc/o	childbirth
toxic/o	poison
ur/o	urine

Combining Forms, Suffixes, Prefixes, and Terminology – Suffixes

Suffix	Meaning
-agon	assemble, gather together
-emia	blood condition
-in, -ine	a substance
-tropin	stimulating the function of
-uria	urine condition

Combining Forms, Suffixes, Prefixes, and Terminology – Prefixes (Slide 1 of 1)

Prefix	Meaning
eu-	good, normal
hyper-	excessive, above
hypo-	deficient, below
oxy-	rapid, sharp, acid
pan-	all
poly-	much or increased
tetra-	four
tri	three

QUICK QUIZ (Slide 3 of 5)

3. Which term means surgical excision of the adrenal gland (specifically)?

- A. Adenectomy
- B. Adrenopathy
- C. Adrenalectomy
- D. Pancreatectomy

QUICK QUIZ (Slide 4 of 5)

4. Which term means a blood condition of too little potassium?

- A. Hyperkalemia
- B. Hypocalcemia
- C. Hypercalciuria
- D. Hypokalemia

Chapter 18

Lesson 18.3

Pathology—Thyroid Gland

Goiter: Enlargement of the thyroid

Hypersecretion

Hyperthyroidism

➤ Graves' disease

- Exophthalmos and proptosis

Hyposecretion

Hypothyroidism

➤ Myxedema

➤ Cretinism

Neoplasms

Thyroid carcinoma

Pathology—Parathyroid

Hypersecretion

Hyperparathyroidism

- Loss of bone density
- Kidney stones
- Hypercalcemia

Hyposecretion

Hypoparathyroidism

- Muscle and nerve weakness
- Tetany
- Hypocalcemia

Pathology—Adrenal Cortex

Hypersecretion

- **Adrenal virilism**

Amenorrhea, hirsutism, acne, voice deepening

- **Cushing syndrome**

Obesity, moonface, thoracic fat deposition

Hyposecretion

- **Addison disease**

Deficient mineralocorticoids and glucocorticoids, hyponatremia, fatigue, weakness, weight loss, low blood pressure

Pathology—Adrenal Medulla

Hypersecretion

- **Pheochromocytoma**

Benign tumor of adrenal medulla

Excess epinephrine and norepinephrine

Hypertension, tachycardia, palpitations, severe headaches, sweating, flushing of the face, and muscle spasms

Pathology—Pancreas

Hypersecretion

- Hyperinsulinism
Hypoglycemia, convulsions, fainting

Hyposecretion

- Diabetes mellitus
Lack of insulin secretion or resistance of insulin in promoting sugar, starch, and fat metabolism in cells
Type 1 : childhood onset, typically; can occur in adulthood
Type 2 : adult onset, typically

Comparison of Type 1 and Type 2 Diabetes

TABLE 18-3 COMPARISON OF TYPE 1 AND TYPE 2 DIABETES MELLITUS		
CATEGORY	TYPE 1*	TYPE 2†
Clinical features	Usually occurs before age 30 Abrupt, rapid onset of symptoms Little or no insulin production Thin or normal body weight at onset Ketoacidosis often occurs	Usually occurs after age 30 Gradual onset; asymptomatic Insulin usually present Obesity in 85% of affected persons Ketoacidosis seldom occurs
Symptoms	Polyuria (glycosuria promotes loss of water) Polydipsia (dehydration causes thirst) Polyphagia (tissue breakdown causes hunger)	Polyuria sometimes seen Polydipsia sometimes seen Polyphagia sometimes seen
Treatment	Insulin	Diet (weight loss); oral hypoglycemics or insulin

*Type 1 formerly was called juvenile (juvenile-onset) diabetes.

†Type 2 formerly was called adult-onset diabetes.

Pathology

Pituitary Gland: (Anterior Lobe)

Hypersecretion

- Acromegaly
- Gigantism

Hyposecretion

- Dwarfism
- Panhypopituitarism

Pathology

Pituitary Gland: (Posterior Lobe)

Hypersecretion

- Syndrome of inappropriate ADH (SIADH)

Excess ADH

Excess water retention

Hyposecretion

- Diabetes insipidus

Deficient ADH

Polyuria and polydipsia

Review Abnormal Conditions of Endocrine Glands

TABLE 18-4 ABNORMAL CONDITIONS OF ENDOCRINE GLANDS

ENDOCRINE GLAND	HYPERSECRETION	HYPOSECRETION
Adrenal cortex	• Adrenal virilism • Cushing syndrome	• Addison disease
Adrenal medulla	• Pheochromocytoma	
Pancreas	• Hyperinsulinism	• Diabetes mellitus
Parathyroid glands	• Hyperparathyroidism (hypercalcemia, osteoporosis, kidney stones)	• Hypoparathyroidism (tetany, hypocalcemia)
Pituitary—anterior lobe	• Acromegaly • Gigantism	• Dwarfism • Panhypopituitarism
Pituitary—posterior lobe	• Syndrome of inappropriate antidiuretic hormone	• Diabetes insipidus
Thyroid gland	• Exophthalmic goiter (Graves disease, thyrotoxicosis) • Nodular (adenomatous) goiter	• Cretinism (children) • Endemic goiter • Myxedema (adults)

Chapter 18

Lesson 18.4

Laboratory Tests

- **Fasting plasma glucose (FPG)**
 - Measures circulating glucose in a patient who has fasted at least 8 hours
- **Serum and urine tests**
 - Measures hormones, electrolytes, glucose, etc., in blood and urine as indicators of endocrine function
- **Thyroid function tests**
 - Measures T3, T4, and TSH in the bloodstream

Clinical Procedures

- Exophthalmometry
- Computed tomography (CT) scan
- Magnetic resonance imaging (MRI)
- Thyroid scan
- Ultrasound examination

Abbreviations (Slide 1 of 3)

Abbreviation	Meaning
A1C	Blood test that measures glycosylated hemoglobin (HbA _{1c}) to assess glucose control
ACTH	Adrenocorticotrophic hormone
ADH	Antidiuretic hormone – vasopressin
Ca+	Calcium
CGMS	Continuous glucose monitoring system
DI	Diabetes insipidus
DKA	Diabetic ketoacidosis
DM	Diabetes mellitus
FBG	Fasting blood glucose
FBS	Fasting blood sugar
FSH	Follicle-stimulating hormone

Abbreviations (Slide 2 of 3)

Abbreviation	Meaning
GH	Growth hormone
GTT	Glucose tolerance test
HbA_{1c}	Test for the presence of glucose attached to hemoglobin; also called AIC
hCG or HCG	Human chorionic gonadotropin
IGF	Insulin-like growth factor
K	Potassium
LH	Luteinizing hormone
MDI	Multiple daily injection
Na	Sodium
OT, OXT	Oxytocin
PRL	prolactin

Abbreviations (Slide 3 of 3)

Abbreviation	Meaning
PTH	Parathyroid hormone (parathormone)
RAI	Radioactive iodine
RIA	Radioimmunoassay
RAIU	Radioactive iodine uptake (imaging test or scan)
SIADH	Syndrome of inappropriate antidiuretic hormone
SMBG	Self-monitoring of blood glucose
STH	Somatotropin – growth hormone
T ₃	Triiodothyronine
T ₄	Thyroxine – tetraiodothyronine
TFT	Thyroid function test
TSH	Thyroid-stimulating hormone

Review Sheet – Combining Forms (Slide 1 of 5)

Combining Form	Meaning
aden/o	gland
adrenal/o	adrenal glands
andr/o	male
calc/o, calici/o	calcium
cortic/o	cortex, outer region

Review Sheet – Combining Forms (Slide 2 of 5)

Combining Form	Meaning
crin/o	secrete
dips/o	third
estr/o	female
gluc/o, glyc/o	sugar
gonad/o	sex glands (ovaries, testes)
home/o	sameness

Review Sheet – Combining Forms (Slide 3 of 5)

Combining Form	Meaning
hormon/o	hormone
kal/i	potassium (an electrolyte)
insulin/o	insulin
lact/o	milk
myx/o	mucus
natr/o	sodium (an electrolyte)

Review Sheet – Combining Forms (Slide 4 of 5)

Combining Form	Meaning
pancreat/o	pancreas
parathyroid/o	parathyroid gland
phys/o	growing
pituitar/o	pituitary gland; or hypophysis
somat/o	body
ster/o	solid structure

Review Sheet – Combining Forms (Slide 5 of 5)

Combining Form	Meaning
thyr/o	thyroid gland
thyroid/o	thyroid gland
toc/o	childbirth
toxic/o	poison
ur/o	urine

Review Sheet – Suffixes

Suffix	Meaning
-agon	assemble, gather together
-emia	blood condition
-in, -ine	a substance
-tropin	stimulating the function of
-uria	urine condition

Review Sheet – Prefixes

Prefix	Meaning
eu-	good, normal
hyper-	excessive, above
hypo-	deficient, below
oxy-	rapid, sharp, acid
pan-	all
tetra-	four
tri-	three

QUICK QUIZ (Slide 5 of 5)

- 5. What is the pathologic condition in which enlargement of the extremities is caused by hypersecretion of the anterior pituitary after puberty?**
- A. Addison disease
 - B. Acromegaly
 - C. Cushing syndrome
 - D. Graves disease